

Gene Sight Eva-Female Infertility NGS Panel

Patient name	: Mrs. XXX	PIN	: XX
Gender/Age	: Female / XX Years	Sample number	: XX
Referring Clinician	: XX	Sample collection date	: NA
Hospital/ Clinic	: XX	Sample received date	: NA
Specimen	: Peripheral blood	Report date	: NA

INDICATION FOR TESTING

Mr. XX and Mrs. XXX were presented with a history of a previous pregnancy genetically diagnosed with noonan's syndrome in the amniotic fluid (reports NA) and subsequently terminated. During their first ART cycle with self-gametes, three? poor quality embryos were formed. Whole exome sequencing in the female partner [XX, 2026] revealed a heterozygous pathogenic variant in *PTPN11*: c.923A>G associated with Noonan Syndrome 1 (AD). Proband, Mrs. XXX has been evaluated for pathogenic variations and polymorphisms and polymorphisms.

RESULTS

FEMALE POLYMORPHISM ANALYSIS

List of polymorphic variants (SNP) identified:

Gene Name	RsID	Variant	Genotype	Zygosity	Observed Phenotype	Recommendations
AMH	rs10407022	c.146G>T	TT	Homozygous Polymorphism	Associated with reduced <i>AMH</i> bioactivity and altered follicular regulation. Lower functional ovarian reserve and reduced responsiveness to mild ovarian stimulation [4-7].	Avoid minimal or low-dose stimulation protocols. Increased FSH dosing and extended stimulation protocols. Adjuvant therapies (DHEA, growth hormone, CoQ10) supplementation.

<i>LHCGR</i>	rs2293275	c.935A>G	GG	Homozygous Polymorphism	Reduced LH/hCG receptor sensitivity. Altered receptor–ligand binding efficiency [6-10].	Higher gonadotropin dose requirement. Recombinant FSH advised. Consider recombinant LH.
<i>MTR</i>	rs1805087	c.2756A>G	GG	Homozygous Polymorphism	Normal to mildly elevated fasting homocysteine. Increased sensitivity to vitamin B12 insufficiency [11-12].	Correlate with serum vitamin B12 and fasting homocysteine levels.
<i>MTHFR</i>	rs1801133	c.665C>T	CT	Heterozygous Polymorphism	Mild reduction in 5-methyltetrahydrofolate production [13-17].	Standard folate supplementation.

FEMALE FERTILITY SINGLE NUCLEOTIDE VARIATION ANALYSIS

No pathogenic or likely pathogenic variant causative of the reported phenotype was detected

*Genetic test results are based on the recommendation of American college of Medical Genetics [1-3].

No other variant that warrants to be reported for the given clinical indication was identified.

Recommendations

- Genetic counseling is recommended.

Methodology: Single Nucleotide Variation - Single Nucleotide Polymorphism

SNV analysis: DNA extracted from the blood, was used to perform targeted gene capture using a custom capture kit (for SNV analysis – 845 genes). The targeted libraries were sequenced to a targeted depth of 80 to 100X using SURFSeq 5000 sequencing platform. This kit has deep exonic coverage of all the coding regions including the difficult to cover regions. The sequences obtained are aligned to human reference genome (GRCh38.p13) using Sentieon aligner and analyzed using Sentieon for removing duplicates, recalibration and re-alignment of indels. Sentieon DNAscope has been used to call the variants. Detected variants were annotated and filtered using the VarSeq software with the workflow implementing the ACMG guidelines for variant classification. The variants were annotated using 1000 genomes (V2), gnomAD (V4.1.0,3.1.2,2.1.1), ClinVar, OMIM, dbSNP, NCBI RefSeq Genes. *In-silico* predictions of the variant were carried out using VS-SIFT, VS-PolyPhen2, PhyloP, GERP++, GeneSplicer, MaxEntScan, NNSplice, PWM Splice Predictor. Only non-synonymous and splice site variants found in the coding regions were used for clinical interpretation. Silent variations that do not result in any change in amino acid in the coding region are not reported.

SNP analysis: Variant analysis (for SNP analysis – 30 genes) and interpretation is done using VarSeq Software. Extensive scientific literature, Information from variant analysis and disease specific databases, population specific research is used to interpret and recommend. All results are finally approved by medical geneticists.

Sequence data attributes

Total reads generated	10.74 Gb
Data ≥ Q30	96.81 %

Genetic test results are reported based on the recommendations of American College of Medical Genetics [1], as described below:

Classification	Interpretation
Pathogenic	A disease-causing variation in a gene which can explain the patients' symptoms has been detected. This usually means that a suspected disorder for which testing had been requested has been confirmed
Likely Pathogenic	A variant which is very likely to contribute to the development of disease however, the scientific evidence is currently insufficient to prove this conclusively. Additional evidence is expected to confirm this assertion of pathogenicity.
Variant of Uncertain Significance	A variant has been detected, but it is difficult to classify it as either pathogenic (disease causing) or benign (non- disease causing) based on current available scientific evidence. Further testing of the patient or family members as recommended by your clinician may be needed. It is probable that their significance can be assessed only with time, subject to availability of scientific evidence.

Disclaimer

- The classification of variants of unknown significance can change over time. Anderson Diagnostics and Labs cannot be held responsible for it.
- Intronic variants, UTR, Promoter region variants and CNV are not assessed using this assay.
- Certain genes may not be covered completely, and few mutations could be missed. Variants not

detected by this assay may impact the phenotype.

- The variations have not been validated by Sanger sequencing.
- The above findings and result interpretation was done based on the clinical indication provided at the time of reporting.
- It is also possible that a pathogenic variant is present in a gene that was not selected for analysis and/or interpretation in cases where insufficient phenotypic information is available.
- Genes with pseudogenes, paralog genes and genes with low complexity may have decreased sensitivity and specificity of variant detection and interpretation due to inability of the data and analysis tools to unambiguously determine the origin of the sequence data in such regions.
- In addition, this test does not cover detection of such structural variants in the genes *SMN1*, *SMN2*, *HBA1*, *HBA2*, *PMS2*, *GBA* and *CYP21A2* which have very high homology. Moreover, the ORF15 transcript isoform (RefSeq id: NM_001034853.1) of the *RPGR* gene associated with retinitis pigmentosa is not sufficiently covered in the test.
- Incidental or secondary findings that meet the ACMG guidelines can be given upon request [3].
- Result interpretation was done based on the literature evidence available at the time of reporting. The clinical significance of the polymorphic variants tested can change over time and Anderson Diagnostics & Labs cannot be held responsible for this.
- This is not a diagnostic test and so not to be considered as diagnosis of any disease. This test is meant only for understanding the polymorphism at a given position and its association with various clinical parameters.

References

1. Richards, S, et al. Standards and Guidelines for the Interpretation of Sequence Variants: A Joint Consensus Recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. *Genetics in medicine: official journal of the American College of Medical Genetics*. 17.5 (2015): 405-424.
2. Amberger J, Bocchini CA, Scott AF, Hamosh A. McKusick's Online Mendelian Inheritance in Man (OMIM). *Nucleic Acids Res*. 2009 Jan;37(Database issue):D793-6. doi: 10.1093/nar/gkn665. Epub 2008 Oct 8.
3. Lee, Kristy et al. "ACMG SF v3.3 list for reporting of secondary findings in clinical exome and genome sequencing: A policy statement of the American College of Medical Genetics and Genomics (ACMG)." *Genetics in medicine: official journal of the American College of Medical Genetics* vol. 27,8 (2025): 101454. doi:10.1016/j.jgim.2025.101454.
4. Rigon, Chiara et al. "Association study of *AMH* and *AMHRII* polymorphisms with unexplained

- infertility." *Fertility and sterility* vol. 94,4 (2010): 1244-1248. doi:10.1016/j.fertnstert.2009.05.025
5. Chen, Di et al. "Can polymorphisms of *AMH/AMHR2* affect ovarian stimulation outcomes? A systematic review and meta-analysis." *Journal of ovarian research* vol. 13,1 103. 4 Sep. 2020, doi:10.1186/s13048-020-00699-4
 6. Udumudi A, Lava KC, Hegde V. High-Risk Genotypes Associated with Poor Response to Controlled Ovarian Stimulation in Indian Women. *J Hum Reprod Sci.* 2023 Jan-Mar;16(1):70-78. doi: 10.4103/jhrs.jhrs_2_23. PMID: 37305771; PMCID: PMC10256945.
 7. Achuri, Aradhya and Anuradha Udumudi. "Improvement of reproductive outcome in hypo-responders using genomics evidence based controlled ovarian stimulation (COS) – Demonstration through case presentations." *Fertility Science and Research* 10 (2023): 145 - 150.
 8. Ga, Ramaraju et al. "Effect of *LHCGR* Gene Polymorphism (rs2293275) on LH Supplementation Protocol Outcomes in Second IVF Cycles: A Retrospective Study." *Frontiers in endocrinology* vol. 12 628169. 11 May. 2021, doi:10.3389/fendo.2021.628169
 9. Lyangasova, Olga V., et al. "*LHCGR* gene variants are associated with ovarian response and miscarriage risk following fresh and frozen embryo transfer." *Научные результаты биомедицинских исследований* 10.3 (2024): 441-456.
 10. Thathapudi, Sujatha et al. "Association of luteinizing hormone chorionic gonadotropin receptor gene polymorphism (rs2293275) with polycystic ovarian syndrome." *Genetic testing and molecular biomarkers* vol. 19,3 (2015): 128-32. doi:10.1089/gtmb.2014.0249
 11. Furness, Denise L F et al. "One-carbon metabolism enzyme polymorphisms and uteroplacental insufficiency." *American journal of obstetrics and gynecology* vol. 199,3 (2008): 276.e1-8. doi:10.1016/j.ajog.2008.06.020
 12. Li, Wen-Xing et al. "Homocysteine Metabolism Gene Polymorphisms (*MTHFR* C677T, *MTHFR* A1298C, *MTR* A2756G and *MTRR* A66G) Jointly Elevate the Risk of Folate Deficiency." *Nutrients* vol. 7,8 6670-87. 10 Aug. 2015, doi:10.3390/nu7085303
 13. Yang, Yi et al. "Association between maternal, fetal and paternal *MTHFR* gene C677T and A1298C polymorphisms and risk of recurrent pregnancy loss: a comprehensive evaluation." *Archives of gynecology and obstetrics* vol. 293,6 (2016): 1197-211. doi:10.1007/s00404-015-3944-2
 14. Turgal, Mert et al. "Methylenetetrahydrofolate Reductase Polymorphisms and Pregnancy Outcome." *Geburtshilfe und Frauenheilkunde* vol. 78,9 (2018): 871-878. doi:10.1055/a-0664-8237
 15. Kasap, Burcu, et al. "Successful treatment of *MTHFR* A1298C and C677T mutant ectopic pregnant woman with extremely high serum β -hCG levels using single dose methotrexate."
 16. Yu, Huan et al. "Severe Adverse Toxic Effects of Low-Dose Methotrexate Treatment on an Ectopic

Pregnancy Patient with Methylenetetrahydrofolate Reductase Mutations: A Case Report.” Frontiers in medicine vol. 8 738315. 18 Nov. 2021, doi:10.3389/fmed.2021.738315

17. Merviel, Philippe et al. “Comparison of two preventive treatments for patients with recurrent miscarriages carrying a C677T methylenetetrahydrofolate reductase mutation: 5-year experience.” The Journal of international medical research vol. 45,6 (2017): 1720-1730. doi:10.1177/0300060516675111

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